

BIF 40I FINAL PAST SOLVE AND MCQS

MCQS

- 1) Fasta Was Develop In (Ans 1988)
- 2) In Corey Paulin Modal Of The Protein Each Atom Is Represented By (Solid Sphere)
- 3) How Many Storages Use For Structural Modling (3)
- 4) Modeler Software Use For Homology Modling Which Use (Python File Format)
- 5) Mass Of The Based Proteomic Begin With Measure Of (Intact Protein Amino Acid)
- 6) Protein Sequence Obtained From (Uni Port)
- 7) During The Formation Of The 5 Nucleotide Hydrogen Bond Energy Release (-12kcal/Mol)
- 8) Which Method Is Not Use For Making Polygenetic??? Msa
- 9) Rna Dont Have Structer??? Quaternary
- 10) Large Numbr Of Sequence Available?? Publically
- 11) Coding Region Of Dna Is??? Exon
- 12) Blast Use For??? Unknown And Known Protein
- 13) Expasy Develop By??? Sib
- 14) Variation In Sequence Develop By??? Insertion, Deletion And Substituion
- 15) Bond Form Between C And N Terminas??? Hydrogen Bond
- 16) X.Y.Z Position Of Protein Are Available By??? Pdb
- 17) Technique Use To Measure The Mass Of Protein??? Mass Spectrometry
- 18) Large Amino Acid??? Tryptophan
- 19) Positive Charge Amino Acid??? Lysine And Arginine
- 20) Mrna Has??? Single Strand
- 21) C.Alpha Atom Are Traced To Recreat??? 3d Structer Of Protein
- 22) Bond Form Between Two Atom??? Energy Release
- 23) Provide Clue Of Precursor Protein Peptide Sequence??? Pst
- 24) Amino Acid Requird For Betasheet??? 3-5
- 25) Are Formed By Bonding Between C And N⁺ Protein Backbone??? Alpha Halix
- 26) Blasta Stand For??? Basic Local Alignment Search Tool
- 27) Amino Acid Have Uniwe Trait Which Can Help In??? Protein Folding
- 28) Align Comparison Between dna/Dna And dna/Protein??? Fasta.
- 29) Ratio Use To Calculate Mass Of Protein??? M/Q.
- 30) Treading Not Required For??? Structer Of Entire Backbone.
- 31) Needle Man Wouch Man Or Smith Water Man Algorithm Use For??? Local And Global Alignmnt.
- 32) Dianamic Programing Recombinant Nucleotide By??? Traseback.
- 33) Insertion And Deletion Cause??? Gap.
- 34) Information Required For Folling Of Amino Acid Is Present In??? Amino Acid.
- 35) Mgf Use For??? Proteomic.
- 36) Nj Algorith Use To Pridict The??? 2nd Structer Of Rna..

Short Questions

I. Seven Steps to Homology Modelling?

- Template recognition and initial alignment • Alignment correction • Backbone generation • Loop modeling
- Side-chain modeling • Model optimization • Model validation

2. Advantage of Folded Protein over Denatured Protein?

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3. Factors That Participate In Folding Protein?

The external factors involved in protein denaturation or disruption of the native state include temperature, external fields (electric, magnetic), molecular crowding, and even the limitation of space, which can have a big influence on the folding of proteins. (Internet)

4. Role of Amino Acid?

Protein plays a crucial role in almost all biological processes and amino acids are the building blocks of it. A large proportion of our cells, muscles and tissue is made up of amino acids, meaning they carry out many important bodily functions, such as giving cells their structure.

5. Role of Global Protein?

Global protein function prediction from protein-protein interaction networks. ... The availability of entire genome sequences and of high-throughput capabilities to determine gene coexpression patterns has shifted the research focus from the study of single proteins or small complexes to that of the entire proteome. (Internet)

6. How We Can Find Loop After Alpha Helix, Beta Sheet?

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7. Acidic and Basic Amino Acid?

There are three amino acids that have basic side chains at neutral pH. These are arginine (Arg), lysine (Lys), and histidine (His). Their side chains contain nitrogen and resemble ammonia, which is a base. Their pKa's are high enough that they tend to bind protons, gaining a positive charge in the process.

8. Points of Bottom up Proteomic?

The major alternative workflow used in high-throughput proteomics is called top-down proteomics and does not use proteolytic digestion. Essentially, bottom-up proteomics is a means of determining the protein make-up of a given sample of cells, tissues, etc. (Internet)

9. Varies RNA Secondary Structure?

Types of RNA Secondary Structures - I • RNA Complementary bases in primary structure bond together to fold RNA into a secondary structure

Types of RNA Secondary Structures - II • RNA 1' sequence (5' – 3') can fold onto itself to make 2' structures.
• Two previous examples of 2' structures include Helix and Hairpin structures.

Types of RNA Secondary Structures - III • RNA can be fold to form helices, bulge loops, and interior loops. An other 2' RNA structure is the Junction or Intersection

10. Objective Base Method and Clustering Approach?

Clustering Approach UPGMA WPGMA Neighbor Joining Single Linkage Complete Linkage

Objective based Methods Least Square Distances Maximum Likelihood Maximum Parsimony

11. Shot Gun and Peptide Mass Finger Printing?

Shotgun proteomics digests the entire protein mix 1st followed by peptide analysis & protein database search

Peptide mass fingerprinting involves protein separation followed by a single protein's peptide analysis.

12. Need For Chou Fasman Algorithms?

The Chou-Fasman algorithm is simple in principle. The conformational parameters for each amino acid were calculated by considering the relative frequency of a given amino acid within a protein, its occurrence in a given type of secondary structure, and the fraction of residues occurring in that type of structure.

13. Substitution and Indels?

Substitution at adjacent nucleotides (primarily substitutions at two adjacent nucleotides, but substitutions at three adjacent nucleotides have been observed.

Indels, being either insertions, or deletions, can be used as genetic markers in natural populations, especially in phylogenetic studies.

14. Advantages and Disadvantages Of Ab Initio Modeling?

Advantages • Ab Initio methods can fold any target sequence using only physical atomic properties • Predictions are mostly accurate and correctly describe the natural folding process.

Disadvantages • Ab initio methods are the very difficult to design (energy function) • These methods are slow due to the huge possibilities • An order of 10^{12} steps are needed to simulate protein folding for medium sized protein structures.

15. PK Value is Less than PH Why?

The lower the pK_a , the stronger the acid and the greater its ability to donate protons. pH depends on the concentration of the solution. This is important because it means a weak acid could actually have a lower pH than a diluted strong acid. On the other hand, the pK_a value is a constant for each type of molecule.

16. What Is Hydrophobicity?

In chemistry, hydrophobicity is the physical property of a molecule that is seemingly repelled from a mass of water. In contrast, hydrophiles are attracted to water. Hydrophobic molecules tend to be nonpolar and, thus, prefer other neutral molecules and nonpolar solvents.

17. Difference Between Folding Protein and Denature Protein, Role Of Folding Protein, and Forces Act on Protein / What Factors Are Used For Folding Proceeds?

Protein folding is the physical process by which a protein chain acquires its native 3-dimensional structure, a conformation that is usually biologically functional, in an expeditious and reproducible manner.

Denaturation of proteins involves the disruption and possible destruction of both the secondary and tertiary structures. Since denaturation reactions are not strong enough to break the peptide bonds, the primary structure (sequence of amino acids) remains the same after a denaturation process.

Protein folding is a process by which a polypeptide chain folds to become a biologically active protein in its native 3D structure. Protein structure is crucial to its function. Folded proteins are held together by various molecular interactions. The amino acid sequence of a protein determines its 3D structure.

The forces involved in protein folding include:

• Electrostatic interactions, • van der Waals interactions, • Hydrogen bonds, • Hydrophobic interactions

18. Prediction of Protein Structure?

Protein structure prediction is the inference of the three-dimensional structure of a protein from its amino acid sequence—that is, the prediction of its folding and its secondary and tertiary structure from its primary structure.

19. How Can We Predict Amino Acid In Alpha Helix And Beta Sheets?

Proteins that convert from alpha helix to beta sheet: implications for folding and disease. The sequence of a protein normally determines which amino acid residues will form alpha helices, and which one beta sheets, to an extent that allows secondary structure prediction to be made with a reasonable reliability.

20. Chou Fasman (Loop Formation after Alpha and Beta Sheet)? / How Loops Are Found After The Finding Of Alpha And Beta Sheets?

Alpha helices and beta strands are connected by these turns and loopsLoops and turns generally lie on the surfaces of proteins so they often participate in interactions between proteins and other molecules. In a loop, there are no regular structures as can be found in helices or beta strands.

21. The Term Expasy Stands For?

Expert Protein Analysis System (Internet)

Expasy • Developed by Swiss Bioinformatics Institute (SIB) • Website provides access to databases and tools • Proteomics, Genomics, Phylogeny, Systems biology, Population genetics, transcriptomics etc. (PPT)

22. Differentiate Between Bottom Up Proteomics And Top Down Proteomics?

Bottom Up Proteomics: Measures peptide masses produced after precursor protein's enzymatic digestion.

Top Down Proteomics: Measures intact proteins followed by their peptides after fragmentation.

23. How Proteins Sequences Identified?

Predicting from DNA/RNA sequences. In biology, proteins are produced by translation of messenger RNA (mRNA) with the protein sequence deriving from the sequence of codons in the mRNA. Predicted protein sequences are an important resource for protein identification by mass spectrometry.

24. Why Folding Is Important?

Proteins have several layers of structure each of which is important in the process of protein folding. The first most basic level of this structure is the sequence of amino acids themselves. The sequencing is important because it will determine the types of interactions seen in the protein as it is folding.

25. Why The Tertiary Structure Of RNA Is Less Stable Over The Secondary Structure Of RNA?

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26. What Is Ab Initio Modelling? Describe The Limitations Of This Model?

Ab initio methods have Anfinsen's thermodynamic hypothesis at the center • These methods attempt to identify the structure with minimum free energy. • Applicable to any sequence • Not very accurate biologically • Accuracy & applicability are limited by our understanding of the protein folding problem.

Limitation • Computationally expensive • Suitable for proteins with less than 100 residues (PPT)

37. Chou Fasman Algorithm?

The Chou–Fasman method is an empirical technique for the prediction of secondary structures in proteins, originally developed in the 1970s by Peter Y. Chou and Gerald D. Fasman.

38. Improvements Of Chou Fasman Algorithm?

The Chou-Fasman algorithm, one of the earliest methods, has been successfully applied to the prediction. ...

We improved Chou-Fasman method in three aspects. (a) Replace the nucleation regions with extreme values of coefficients calculated by the continuous wavelet transform.

39. Limitations Of Chou Fasman?

The Chou–Fasman method has been widely used for predicting protein secondary structure. It is based on knowledge of the potential of amino acid residues to form α -helical or β -sheet regions in proteins. Our main interest in this study was to examine the reliability of these Chou–Fasman parameters.

40. Scoring Formulas?

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41. Beta Sheet By Chou Fasman and Alpha Helix By Chou fasman?

The Chou–Fasman method predicts helices and strands in a similar fashion, first searching linearly through the sequence for a "nucleation" region of high helix or strand probability and then extending the region until a subsequent four-residue window carries a probability of less than 1.

The Chou–Fasman method is an empirical technique for the prediction of secondary structures in proteins, originally developed in the 1970s by Peter Y. Chou and Gerald D. Fasman. The method is based on analyses of the relative frequencies of each amino acid in alpha helices, beta sheets, ...

42. Propensity?

In the statistical analysis of observational data, propensity score matching (PSM) is a statistical matching technique that attempts to estimate the effect of a treatment, policy, or other intervention by accounting for the covariates that predict receiving the treatment.

43. Fragmentation Of Protein?

Protein fragmentation is a common practice in the study of protein structure and function. 1'2 Specific cleavage of certain peptide bonds in the primary structure of a protein is essential in.

44. Hydrophobic Amino Acids?

Hydrophobic Amino Acids. Amino acids are grouped according to what their side chains are like. The nine amino acids that have hydrophobic side chains are glycine (Gly), alanine (Ala), valine (Val), leucine (Leu), isoleucine (Ile), proline (Pro), phenylalanine (Phe), methionine (Met), and tryptophan (Trp).

45. Optimal Energy Function In Protein?

Results: For a single protein sequence, the probability of success (i.e. the probability that the folded state is the lowest energy state) is derived. We then maximize the average probability of success for a set of proteins to obtain the optimal potential energy function.

46. Blast And Fasta Tools In Sequence Alignment And Its Differences?

BLAST is the most widely used tool for the local alignment of nucleotide and amino acid sequences. FASTA is a fine similarity searching tool which uses sequence patterns or words. ... The main difference between BLAST and FASTA is in the similarity searching strategies used in each tool.

47. Protein Scoring?

Protein digestibility-corrected amino acid score (PDCAAS) is a method of evaluating the quality of a protein based on both the amino acid requirements of humans and their ability to digest it.

48. Why Is It Difficult To Predict The Structure Of Protein?

It is much more difficult to compare two models that predict a protein whose structure is unknown and determine which one is more accurate." ... Many protein structures are unknown. Groups around the world are all attempting to determine the structures of proteins that are important for current research.

49. Experimental Techniques In Bioinformatics?

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50. Silico Fragments Comparison?

Matching experimental fragments with in silico fragments is the final resort in protein search & identification

51. How Edman Degradation Work?

Edman degradation. Edman degradation, developed by Pehr Edman, is a method of sequencing amino acids in a peptide. In this method, the amino-terminal residue is labeled and cleaved from the peptide without disrupting the peptide bonds between other amino acid residues.

52. Which Online Tool Is Used To Predict RNA?

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53. Example About The Chou Fasman Secondary Structure Of Protein?

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54. Integrative Scoring Scheme?

Three individual scores can be obtained including 1. MW Match Score 2. PST Match Score 3. In vitro & In silico Match Score.

55. How Propensity Table Help Us?

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56. Distinguish In Multiple Sequence Alignment And Pairwise Sequence Alignment?

Multiple Sequence Alignment:

A multiple sequence alignment is a sequence alignment of three or more biological sequences, generally protein, DNA, or RNA. In many cases, the input set of query sequences are assumed to have an evolutionary relationship by which they share a linkage and are descended from a common ancestor.

Pairwise Sequence Alignment:

Pairwise Sequence Alignment is used to identify regions of similarity that may indicate functional, structural and/or evolutionary relationships between two biological sequences (protein or nucleic acid).

57. What Are Uses Of Mass Spectrometry?

Mass spectrometry (MS) is an analytical technique that ionizes chemical species and sorts the ions based on their mass-to-charge ratio. In simpler terms, a mass spectrum measures the masses within a sample. Mass spectrometry is used in many different fields and is applied to pure samples as well as complex mixtures.

58. How We Can Get Scoring In BUP Proteomics?

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59. About Prosite Pattern?

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60. Define Tblastx blastn blast?

tblastx (translated BLAST)

blastn (translated BLAST)

BLAST performs searches by quick alignments on sequences.

61. Scoring In Homology Modelling?

The scoring functions (1) and (2) are sequence-based: score (1) compares the secondary structure prediction for the target sequence with the actual secondary structure assignment of the template protein, while score (2) evaluates the sequence similarity between the target and template sequence.

62. Why Model Of Protein Structure Is Necessary?

Proteins are chains of amino acids joined together by peptide bonds. Many conformations of this chain are possible due to the rotation of the chain about each C α atom. It is these conformational changes that are responsible for differences in the three dimensional structure of proteins.

63. How Positive And Negative Energy Relapsed In A Molecule?

The change in enthalpy is negative in exothermic processes, because energy is released from the system into its surroundings. Generally, a positive change in enthalpy is required to break a bond, while a negative change in enthalpy is accompanied by the formation of a bond.

64. Enlist Few Points In Scope Of Bioinformatics?

The scope of a bioinformatics project (or lab) varies widely depending on the balance of math/stats, computational/software/engineering, and biological/molecular training of the researchers involved and the data or subject matter they deal with particularly DNA sequences, gene expression, protein, epigenetics.

65. Different Secondary Structures Of RNA?

The secondary structures of biological DNA's and RNA's tend to be different: biological DNA mostly exists as fully base paired double helices, while biological RNA is single stranded and often forms complex and intricate base-pairing interactions due to its increased ability to form hydrogen bonds stemming from the.

66. Silico And Vitro?

In silico is an expression meaning "performed on computer or via computer simulation" in reference to biological experiments.

For any jth residue in sequence, we calculate $f(\text{Total}) = f(j) f(j+1) f(j+2) f(j+3)$ (tetra peptide).

If,

$f(\text{Total}) > 0.000075$.

The average value for $P(\text{turn}) > 1.00$ in the tetra peptide.

The averages for the tetra peptide are such $P(\alpha\text{-helix}) < P(\text{turn}) > P(\beta\text{-sheet})$,

It is a Turn!

Advantage and disadvantage of Ab initio?

Advantages:

- Ab Initio methods can fold any target sequence using only physical atomic properties.
- Predictions are mostly accurate and correctly describe the natural folding process.

Disadvantages:

- Ab initio methods are the very difficult to design (energy function).
- These methods are slow due to the huge possibilities.

Bioinformatics role?

An interdisciplinary field

New but rapidly developing field

Low requirement on infrastructure and research equipment

Vast opportunities for scientific discovery

PAM matrix?

PAM means "Point Accepted Mutations"

Point accepted mutation is a substitution of one amino acid by another such that the protein functions stays conserved.

PAM unit is a time in which about 1% of amino acids in a sequence undergo accepted mutations.

PAM matrices are scoring matrices that are useful in computing sequence alignment scores.

CLUSTALW?

Performs alignment in slow/accurate or fast/approximate

Create multiple alignments, optimize existing alignments, profile analysis, and create phylogenetic trees.

CLUSTALW can use multiple file formats including EMBL/SwissProt and Pearson (Fasta) etc

Function of Tandem MS?

Tandem MS helps in measurement of mass to the fragments as well. This process provides another step in further scoring and ranking and Protein identification thus becomes easier.

Steps present in flow chart of homology modeling?

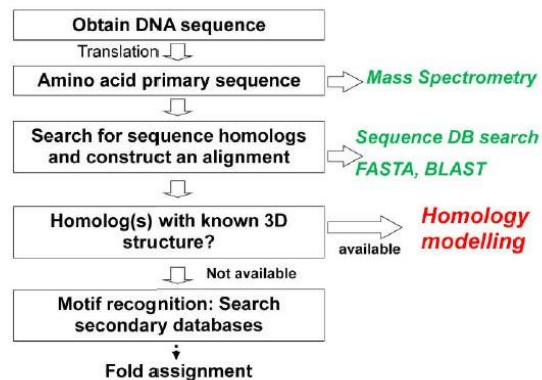
1. Template recognition and initial alignment.
2. Alignment correction.
3. Backbone generation.
4. Loop modeling.
5. Side-chain modeling.
6. Model optimization.
7. Model validation

How homology modeling is used for knowing the sequence of unknown protein sequence?

If another protein which has a similar sequence also has its structure known, the structure of an unknown protein can be predicted based on that similar protein. So, it is then possible to identify unknown protein structures by just examining the homologous protein sequences. Good sequence alignment and identity

ensures that homology modelling will give accurate results. Thus, Homology modeling is used to predict structures of proteins having high sequence similarity with other proteins with known structures:

Workflow of Structural Modelling



Silico fragment scoring?

- Count the matches between in silico and in vitro peaks.
- Give an equivalent score to the candidate protein.
- Weigh each of the aforementioned match by the mass error.
- Accumulate the score

How we can get scoring in BUP proteomic?

There are two approaches to perform BUP:

- Peptide Mass Fingerprinting.
- Shotgun Proteomics

ProSight pattern:

Kelleher et al have developed an Engine i.e. Prosight PTM

ProSight PTM is the state of the art in top down proteomics search.

Using Prosight PTM, post-translational modifications can be accurately identified.

Top down proteomics:

Measures intact proteins followed by their peptides after fragmentation Score in Protein Fragmentation:- If we can:

Measure the mass of fragments using MS.

Calculate the Protein Fragmentation Techniques theoretical mass of the fragments. Then, we can award score on the basis of the similarity of experimental and theoretical mass.

Molecular biology?

It is the branch of biology that deals with the structure and function of the macromolecules (e.g. proteins and nucleic acids) essential to life. Molecular biology is the study of biology at a molecular level.

Protein Fragmentation techniques?

- Electron Capture Dissociation (ECD).
- Electron Transfer Dissociation (ETD).
- Collision Induced Dissociated (CID).

Types of secondary structures of RNA:

I. Single stranded:

- 3' end may fold on to the 5' end.

2. Helices:

- Double stranded RNA helix of stacked base pairs

3. Hairpin loop:

- The loop of the hairpin must at least four bases long to avoid steric hindrance with base-pairing in the stem part of the structure

4. Bulge Loops:

- Bulges, are formed when a double-stranded region cannot form base pairs perfectly.

5. Interior loop:

- Interior loops are formed by an asymmetric number of unpaired bases on each side of the loop.

6. Junctions or intersections:

- Junctions include two or more double-stranded regions converging to form a closed structure.
- The unpaired bases appear as a bulge.

Rooted Un Rooted Tree?

Rooted Tree:

- Each node with descendants represents the inferred most recent common ancestor of the descendants.
- The edge lengths in some trees may be interpreted as time estimates.
- Rooted trees can show temporal evolutionary direction.
- Expensive.

Unrooted:

- Only the relatedness of the leaf nodes.
- Do not require the ancestral root to be known or inferred.
- Less expensive.

Uses:

In bioinformatics, such as

- Rooted and unrooted trees can be used to show phylogenetic relationships between sequences.